

## COMPILER DESIGN - QUIZ 4 (SET B)

Student id:

Name:

Section:

$S \rightarrow id = E ; \{ gen(top.get(id.lexeme) '=' E.addr); \}$

$| L = E ; \{ gen(L.array.base '[' L.addr ']' '=' E.addr); \}$

$E \rightarrow E_1 + E_2 \{ E.addr = new Temp(); gen(E.addr '=' E_1.addr '+' E_2.addr); \}$

$| id \{ E.addr = top.get(id.lexeme); \}$

$| L \{ E.addr = new Temp(); gen(E.addr '=' L.array.base '[' L.addr ']); \}$

$L \rightarrow id [ E ] \{ L.array = top.get(id.lexeme); L.type = L.array.type.elem; L.addr = new Temp();$   
 $gen(L.addr '=' E.addr '*' L.type.width); \}$

$| L_1 [ E ] \{ L.array = L_1.array; L.type = L_1.type.elem; t = new Temp(); L.addr = new Temp();$   
 $gen(t '=' E.addr '*' L.type.width); gen(L.addr '=' L_1.addr '+' t); \}$

Using the grammar and semantic rules given, **draw** the annotated parse tree and **write** the generated Three Address Code for the following statement: [ 10 Marks ]

$$B[i][j] = C[k] + a$$

Assume that both B and C are **integer** arrays. **The dimension of B is 4 x 5 while the array B is one dimensional with size 3.** Each integer occupies **4 bytes**. Variables i, j and k are integers. The gen() function generates strings and the Temp() class creates a temporary variable which is **‘t’ with a subscript 1, 2, 3 and so on.**

|                             |                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------|
| <b>B → B1 &amp;&amp; B2</b> | B1.true =<br>B1.false =<br>B2.true =<br>B2.false =<br>B.code = B1.code       B2.code |
|-----------------------------|--------------------------------------------------------------------------------------|

**Complete the SDD for this boolean expression. [ 5 Marks]**